

ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

NER AND HUMAN ANNOTATIONS

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Classic Named Entity Recognition (NER)

Some questions to think about:

- Can massive raw texts help? → Neural language model
- Can dictionaries help? → Distant supervised setting
- **Are human annotations always correct?**
- Is Tokenizer always good?

State-of-the-art F1 score on this test set is already around **93%**

~**5.38%** test sentences have annotation mistakes

- Significant amount!

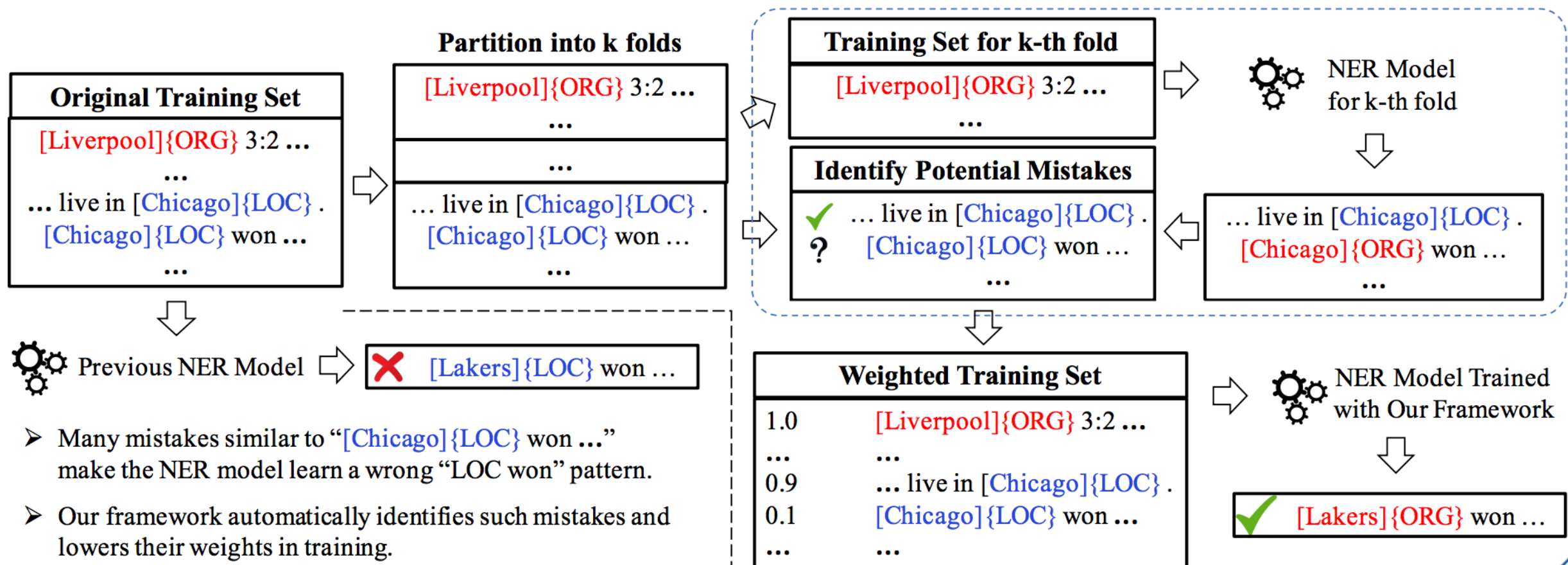
Sentence	Original labels	Corrected labels
Sporting Gijon 15 4 4 7 15 22 16	[Sporting]{ORG}	[Sporting Gijon]{ORG}
SOCCER - JAPAN GET LUCKY WIN , CHINA IN SURPRISE DEFEAT .	[JAPAN]{LOC}, [China]{PER}	[JAPAN]{LOC}, [China]{LOC}
NZ 's Bolger says Nats to meet NZ First on Sunday .	[NZ]{LOC}, [Bolger]{PER}, [Nats]{PER}, [NZ]{LOC}	[NZ]{LOC}, [Bolger]{PER}, [Nats]{ORG}, [NZ First]{ORG}
Seagramd ace 20/11/96 5,000 Japan	[Seagramd] {MISC}, [Japan]{LOC}	[Seagramd ace]{MISC}, [Japan] {LOC}

Higher F1 score with smaller variance

- Better reflects the real performance
- This corrected test set should be adopted in future research

Method	Original	Corrected
LSTM-CRF	90.64 (± 0.23)	91.47 (± 0.15)
LSTM-CNNs-CRF	90.65 (± 0.57)	91.87 (± 0.50)
VanillaNER	91.44 (± 0.16)	92.32 (± 0.16)
Elmo	92.28 (± 0.19)	93.42 (± 0.15)
Flair	92.87 (± 0.08)	93.89 (± 0.06)
Pooled Flair	93.14 (± 0.14)	94.13 (± 0.11)

CROSSWEIGH (EMNLP'19): HANDLE NOISY TRAINING SET



Random Partition may be ineffective

- Neural NER models will overfit the annotation mistakes observed during training

Entity Disjoint Filtering

- In each fold, if a “training” sentence contains any entities appeared in the “testing” set, it will be discarded during the “training”

CrossWeigh is effective with many NER models

	Original CoNLL03		Corrected CoNLL03	
	w/o CrossWeigh	w/ CrossWeigh	w/o CrossWeigh	w/ CrossWeigh
VanillaNER	91.44 (± 0.16)	91.78 (± 0.06)	92.32 (± 0.16)	92.64 (± 0.08)
Flair	92.87 (± 0.08)	93.19 (± 0.09)	93.89 (± 0.06)	94.18 (± 0.06)
Pooled-Flair	93.14 (± 0.14)	93.43 (± 0.06)	94.13 (± 0.11)	94.28 (± 0.05)

Entity Disjoint Filtering is important

Method	Original	Corrected
w/o CrossWeigh	92.87 (± 0.08)	93.89 (± 0.06)
w/ CrossWeigh	93.19 (± 0.09)	94.18 (± 0.06)
– Entity Disjoint	92.88 (± 0.11)	93.84 (± 0.08)
+ Random Discard	93.01 (± 0.10)	93.94 (± 0.10)

Twitter & Low-resource

Dataset	w/o CrossWeigh	w/ CrossWeigh
WNUT'17	48.96 (± 0.97)	50.03 (± 0.40)
Sinhalese	66.34 (± 0.34)	67.68 (± 0.21)

CROSSWEIGH: IDENTIFY ANNOTATION MISTAKES

- CoNLL03 train, dev & test as a super training set
- Apply CrossWeigh to identify annotation mistakes on the test set
- Evaluate against 186 human corrections
 - Almost 80% of mistakes can be detected

Algorithm	Potential Mistakes	Actual Mistakes	Precision	Recall	F1
Flair	573.0	144.0	0.2513	0.7742	0.3794
VanillaNER	821.67	146.33	0.1781	0.7867	0.2904

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Some questions to think about:

- Can massive raw texts help? → Neural language model
- Can dictionaries help? → Distant supervised setting
- **Are human annotations always correct? → Auto-**

Correction

- Is Tokenizer always good?